

Laura Brustenga i Moncusí | Mathematician

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Profile

Mathematician with a PhD in algebraic geometry from the Universitat Autònoma de Barcelona (2020) and five peer-reviewed publications. My work has ranged from pure algebraic geometry to reaction-network theory and numerical algebraic geometry. I have taught mathematics and statistics at university level for eight years, and I build software: Julia packages with tests, continuous integration and documentation, and an Android app. Based in Barcelona.

Experience

Universitat Autònoma de Barcelona

Department of Mathematics

Substitute Professor

2024–

Statistics for Telecommunications Engineering; Linear Algebra for Mathematics and Telecommunications Engineering; Discrete Mathematics seminars. Active-learning methods and digital teaching materials.

Universitat Autònoma de Barcelona

Academic Advisor

2023–

Individual academic guidance for undergraduate mathematics students: study strategies and career planning.

University of Copenhagen

Department of Mathematical Sciences, Mathematical Biology group

Postdoctoral Researcher

2020–2021

Real algebraic geometry methods for parametric reaction networks, funded by the Novo Nordisk Foundation, mentored by Elisenda Feliu. Implemented algorithms for the steady-state analysis of biologically motivated dynamical systems; the work produced the paper *Disguised toric dynamical systems*.

2018–2019: Research visits: KTH Royal Institute of Technology, Stockholm (April–June 2018, with Roy Skjelnes); Max Planck Institute for Mathematics in the Sciences, Leipzig (March–May 2019, Nonlinear Algebra group, with Bernd Sturmfels); Technische Universität Berlin (September–December 2019, Einstein Foundation project *Nonlinear Algebra*, with Michael Joswig and Bernd Sturmfels).

Barcelona Graduate School of Economics

Research Assistant

2014–2015

Extended the exercise database of ACME, a digital platform for personalised learning and assessment at universities. Maxima programming and exercise design.

Universitat Autònoma de Barcelona

Department of Mathematics

Research Assistant

2010–2011

Built a web tool generating proposals for adapting student plans from pre-Bologna diplomas to the Bologna-era Statistics degree across Catalan universities.

Education

Universitat Autònoma de Barcelona

PhD in Mathematics

2016–2020

Thesis *Parametrising clusters of sections*, supervised by Joaquim Roé, funded by a PIF predoctoral fellowship. *Cum Laude*, with International Doctorate Mention. Thesis · defence slides

Universitat Autònoma de Barcelona

Master in Advanced Mathematics

2010–2013

Universitat Autònoma de Barcelona

Llicenciatura in Mathematics

2005–2010

Five-year programme.

Interuniversity programme (UAB, UPC, UB, UPF, UOC)

Barcelona

Master in Teacher Training (Mathematics)

2013–2014

Pedagogical and didactic training for secondary education.

Publications

Peer-reviewed:

1. *Disguised toric dynamical systems*, with G. Craciun and M.-Ş. Sorea. *Journal of Pure and Applied Algebra* **226** (2022), no. 8.
2. *The Waring rank of binary binomial forms*, with S. K. Masuti. *Pacific Journal of Mathematics* **313** (2021), no. 2.
3. *Inverting catalecticants of ternary quartics*, with E. Cazzador and R. Homs. *Le Matematiche* **76** (2021), no. 2.
4. *96120: The degree of the linear orbit of a cubic surface*, with S. Timme and M. Weinstein. *Le Matematiche* **75** (2020), no. 2.
5. *On the universal scheme of r -relative clusters of a family*. *Communications in Algebra* **45** (2017), no. 6. Published under an earlier name.

Preprints:

6. *The blow up split sections family*. arXiv:1808.03062
7. *On the Waring rank of binary forms: the binomial formula and a dihedral cover of rank two forms*, with S. K. Masuti. arXiv:1901.08320

Links: DOIs, arXiv identifiers and BibTeX for everything above are on laurabmo.com/research.

Software

- **SymbolicCRN.jl** — author and maintainer. A registered Julia package for symbolic analysis of chemical reaction networks, parametrising steady states by convex parameters; it bridges Catalyst.jl and OSCAR. Continuous integration, tests, documentation, tagged releases, MIT licensed.
- **BudanTables.jl** — Julia package for plotting Budan's tables of polynomials, with pluggable root-finding backends.
- **PKPDMModels.jl** — Julia package for pharmacometric modelling.
- **SoftReminder** — an Android application in Kotlin, designed and built independently. laurabmo.com/softreminder

Teaching

University: Eight years at the Universitat Autònoma de Barcelona and the University of Copenhagen. Courses: Statistics, Linear Algebra, Algebraic Geometry, Biostatistics, Discrete Mathematics.

Copenhagen: Teaching assistant, *Introduction to Applied Algebra and Geometry* (Master, autumn 2021), *Algebraic Geometry I* (Master, spring 2021), *Mathematics in Spanish* (Bachelor, autumn 2020).

UAB: Statistics and Biostatistics teaching assistant, autumn 2016 to autumn 2018: Statistics for Environmental Science, Biostatistics for Biotechnology, Algebra.

PhD course: *Algebraic Geometry for Molecular Biology*, Max Planck Institute Leipzig, summer 2022.

Other: Secondary-school mathematics teacher, IES Gerbert d'Aurillac, Barcelona, 2015–2016. Private tutoring in mathematics, physics and English from 2015.

Training: *Introduction to University Pedagogy* (University of Copenhagen, 2021), *Digital Teaching Tools* (Barcelona Education Consortium, 2024), *Inclusive Teaching Practices* (AQU, 2023).

Selected talks

2022: *96120: The degree of the linear orbit of a cubic surface* — Congreso Bienal de la Real Sociedad Matemática Española, Ciudad Real

2021: *Multistationarity in chemical reaction networks* — Sage/Oscar Days for Combinatorial Algebraic Geometry, ICERM, Brown University. *Pseudomonas aeruginosa: virulence vs biofilm formation* — SIAM

2020: *The blow up split sections family* — V Congreso de Jóvenes Investigadores RSME

2019: *Clusters of sections for smooth families* — 2nd BMS-BGSMath Junior Meeting, Zuse Institute Berlin. *Relative clusters for smooth families* — Congreso Bienal de la RSME, Santander. *Numerically computing the local dimension of an algebraic set* — Seminar on Discrete and Convex Geometry, TU Berlin

2018: *Waring rank for forms: binary binomials and some enumerative geometry* — Ideals of Powers and Powers of Ideals, Politecnico di Torino

Full list: Talks, seminars and posters are listed at laurabmo.com/research.

Outreach

- Dissemination articles in *SCM/Notícies*, the magazine of the Societat Catalana de Matemàtiques, with M. Prats Soler (2020–2022).
- Workshop coordinator, *Bojos per les matemàtiques*, Universitat Autònoma de Barcelona, 2018–2022. Co-organiser of *Xperimentant les matemàtiques* for the International Day of Women and Girls in Science, Institut d'Estudis Catalans, 2018.

Technical skills

Programming: Julia (primary), Python (NumPy, SciPy, Jupyter), Kotlin, C++, Emacs Lisp, R, SQL, L^AT_EX.

Scientific: OSCAR, Singular, Macaulay2, polymake, GAP, Catalyst.jl, DifferentialEquations.jl, HomotopyContinuation.jl, MATLAB, Maxima, Octave.

Engineering: Git, GitHub Actions, unit testing, package and API design, technical documentation, Docker Compose, GNU/Linux.

AI tooling: Claude Code, Model Context Protocol (MCP) servers, agentic workflow design.

Languages

Catalan, Spanish: Native.

English: C1/C2, academic and professional.

Italian: Basic.